

BillieBot — Sensor Specification Reference Sheet

Revision: 1.0 **Date:** 2026-08-02 **Prepared by:** Systems / Hardware Engineering **Purpose:** Single-page reference for the specifications of all BillieBot sensors, with emphasis on measurement performance, electrical load, digital interface, and the structure/units of the data each device returns.

Legend — Value Provenance

Every number in this document carries one of the following markers. **Unmarked values are manufacturer-stated** (datasheet, official product brief, or official vendor documentation).

Marker	Meaning
<i>(no marker)</i>	Manufacturer-stated value from a datasheet or official vendor documentation
[CALC]	Derived arithmetically by the author from manufacturer-stated values. The derivation is shown.
[EST]	Estimated. Not published by the manufacturer; based on community measurement, engineering judgment, or a comparable part. Verify on hardware before relying on it.
[CONFLICT]	Two official sources disagree. Both values are shown. Requires bench verification.

1. Sensor Suite at a Glance

#	Sensor	Primary Measurand(s)	Interface	Supply	Typ. Power
1	Luxonis OAK-D Lite (Auto-Focus)	RGB image, stereo depth, 6-axis inertial, on-device NN inference	USB-C (USB 2/3)	5 V (bus)	4–5 W
2	Adafruit MLX90640 (55°), PID 4407	Scene temperature (32×24 thermal array)	I ² C (0x33)	3–5 V in / 3.3 V core	~60 mW [CALC]
3	RPi Camera Module 3 NoIR (75°), SC0873	Visible + near-IR image	MIPI CSI-2 + I ² C	3.3 V (CSI)	~0.8 W [EST]
4	Seeed reSpeaker XVF3800 USB 4- Mic Array	Audio, direction of arrival, voice activity	USB-C (UAC 2.0) + I ² C	5 V (bus)	~1–2 W [EST]
5	DFRobot Gravity 10 DOF AHRS, SEN0253	Orientation, accel, angular rate, magnetic field, pressure, temperature	I ² C (0x28, 0x76)	3.3–5 V	~40–60 mW [CALC]

Bus Allocation Summary

- **USB:** 2 devices — OAK-D Lite and reSpeaker XVF3800. See §7.2 for a critical current-budget warning on the OAK-D Lite.
- **I²C:** 3 devices — MLX90640 (0x33), BNO055 (0x28), BMP280 (0x76). **No address conflicts.** However, see §7.3 for a bus-speed constraint that argues for splitting the MLX90640 onto its own bus.
- **MIPI CSI-2:** 1 device — Camera Module 3 NoIR. Consumes one CSI port on the host.

2. Luxonis OAK-D Lite — Auto-Focus Variant

Part number: OAK-D-LITE-AF (Luxonis SKU A00483) **Architecture:** Intel Movidius Myriad X VPU (Luxonis RVC2 platform) **Physical:** 91 × 28 × 17.5 mm; aluminium housing with Gorilla Glass front; ¼-20 tripod mount; VESA 75 mm M4 mounting holes

This is a **multi-output device**. It presents five distinct data streams to the host, covered in §2.1 through §2.5.

2.0 Device-Level Electrical & Environmental

Parameter	Value	Notes
Supply	5 V nominal via USB-C VBUS	5.25 V max recommended; 3.5–5.5 V absolute
Max input current (abs max)	1.5 A	Absolute maximum rating
Power consumption (min/typ/max)	4 W / 5 W / 6 W	Recommended operating conditions
Idle power (VPU booted, no pipeline)	2.5 W	
Typical operating current @ 5 V	~0.8–1.0 A [CALC]	$4\text{--}5\text{ W} \div 5\text{ V}$
Peak operating current @ 5 V	~1.2 A [CALC]	$6\text{ W} \div 5\text{ V}$
Idle current @ 5 V	~0.5 A [CALC]	$2.5\text{ W} \div 5\text{ V}$
Ambient operating temperature	–20 °C to +50 °C	At full VPU utilization
VPU thermal shutdown	105 °C junction	DepthAI shuts the device down automatically
Host interface	USB-C, USB 2.0 / USB 3	Luxonis docs cite up to 10 Gbps; datasheet cites USB 3.1 Gen1
On-board EEPROM	32 kb, I ² C	Stores factory calibration

Power breakdown by subsystem (Luxonis):

Subsystem	Contribution
Base + camera streaming	2.5 – 3 W
AI (neural inference) subsystem	up to 1 W
Stereo depth pipeline	up to 0.5 W
Video encoder	up to 0.5 W

Design note: Reducing pipeline FPS reduces power, because subsystems are no longer saturated.

2.1 Output A — RGB Color Camera

Parameter	Value
Image sensor	Sony IMX214
Nominal resolution	13.1 MP; 4208×2160 active... 4208×3120 @ up to 60 fps (sensor-level)
Optical format	1/3.06" (Luxonis docs) / 1/3.1" (datasheet) [CONFLICT] — negligible practical difference
Field of view (D / H / V)	$81^\circ / 69^\circ / 54^\circ$ (Luxonis docs); datasheet states DFOV 81.3°
Shutter type	Rolling (electronic)
Focus	Auto-focus, 8 cm $\rightarrow \infty$; software-settable to a fixed lens position via DepthAI API
Recommended close-range use	8 cm - 50 cm (this is the reason to choose AF over FF)
IR sensitivity	No — IR cut filter fitted
Hardware encoding	H.264, H.265, MJPEG — 4K @ 30 fps, 1080p @ 60 fps

Data received by host:

- **Uncompressed:** `ImgFrame` messages. Default color order is **NV12** (YUV 4:2:0 semi-planar, 8-bit). Also selectable: YUV420p, BGR/RGB 8-bit interleaved or planar, GRAY8. Each frame carries a monotonic sequence number and a device-synchronized timestamp.
- **Compressed:** H.264 / H.265 elementary stream or MJPEG frames as `ImgFrame` payloads.
- **Bandwidth, 1080p30 NV12:** $1920 \times 1080 \times 1.5 \text{ bytes} \times 30 \text{ fps} \approx \mathbf{93 \text{ MB/s}}$ **[CALC]** — plan for USB 3 if streaming uncompressed color at this rate.

2.2 Output B — Stereo Mono Cameras (Left / Right)

Parameter	Value
Image sensor (×2)	OmniVision OV7251
Active pixels	640 × 480 @ up to 120 fps
Optical format	1/7.5"
Field of view (D / H / V)	86° / 73° / 58° (Luxonis docs); datasheet states 85.6° / 72.9° / 57.7°
Shutter type	Global — important for a moving platform; no rolling-shutter skew
Focus	Fixed, 6.5 cm → ∞
Native output format	8-bit or 10-bit RAW
IR sensitivity	No (per datasheet); note the OAK-D Lite has no IR dot projector and no IR illuminator

Data received by host: `ImgFrame` messages, typically **GRAY8** (8-bit luminance), 640 × 480, one stream per camera. **Bandwidth, both cameras @ 30 fps:** 640 × 480 × 1 byte × 30 × 2 ≈ **18.4 MB/s [CALC]**

2.3 Output C — Stereo Depth

Parameter	Value
Stereo baseline	75 mm
Ideal depth range	~80 cm – 12 m (Luxonis hardware docs) [CONFLICT]
Ideal depth range (alternate)	40 cm – 8 m (Luxonis shop page) [CONFLICT]
MinZ, 480P + extended disparity	~20 cm
MinZ, 480P standard	~35-40 cm
Median depth accuracy, < 4 m	< 2 % absolute depth error
Median depth accuracy, 4 – 7 m	< 4 % absolute depth error
Median depth accuracy, 7 – 10 m	< 6 % absolute depth error
Max disparity throughput	200+ fps (stereo pipeline capability)

[CONFLICT] — action required: Luxonis' own hardware documentation and store page state materially different "ideal range" figures (80 cm–12 m vs. 40 cm–8 m). The difference stems from which mono resolution and disparity mode is assumed.

Characterize the actual usable range on BillieBot in its final configuration before fixing navigation or obstacle-avoidance thresholds.

Data received by host:

- **Depth map:** `ImgFrame`, 640×480 , **uint16, 1 LSB = 1 mm** by default. A value of `0` means *invalid / no measurement*, not zero distance — filter these explicitly.
- **Disparity map:** `ImgFrame`, uint8 (or uint16 with subpixel enabled). Convert to depth via $Z = (\text{baseline_mm} \times \text{focal_length_px}) / \text{disparity}$.
- **Confidence map:** optional uint8 per-pixel confidence.
- **Point cloud:** derivable on-device or host-side; XYZ in millimetres, optionally colored from the RGB camera (RGB-depth alignment is supported).
- **Bandwidth, 640×480 depth @ 30 fps:** $640 \times 480 \times 2 \text{ bytes} \times 30 \approx \mathbf{18.4 \text{ MB/s}}$ [CALC]

Post-processing available on-device: median filter, speckle filter, temporal filter, spatial filter, decimation, threshold filter, RGB-depth alignment.

2.4 Output D — On-Board IMU (Bosch BMI270)

⚠ CRITICAL PROCUREMENT NOTE: Luxonis states that **all OAK-D-Lite units from the original Kickstarter campaign shipped WITHOUT an IMU** as a cost-reduction measure. Retail units have the BMI270. **Verify IMU presence on the specific unit used in BillieBot** by enumerating the device before designing this stream into the architecture.

Parameter	Value
Device	Bosch BMI270, 6-axis (3-axis accelerometer + 3-axis gyroscope)
Magnetometer	None
On-sensor fusion (quaternion / rotation vector)	Not available on the BMI270 path in DepthAI
Internal connection	SPI (VPU ↔ IMU); delivered to host over USB

Accelerometer:

Parameter	Value
Full-scale options	$\pm 2 / \pm 4 / \pm 8 / \pm 16$ g
Sensitivity	16384 / 8192 / 4096 / 2048 LSB/g respectively
Noise density (typ)	0.16 mg/ $\sqrt{\text{Hz}}$
Native ODR range	12.5 Hz – 1600 Hz

Gyroscope:

Parameter	Value
Full-scale options	$\pm 125 / \pm 250 / \pm 500 / \pm 1000 / \pm 2000$ °/s
Sensitivity	262.144 / 131.072 / 65.536 / 32.768 / 16.384 LSB/(°/s) respectively
Noise density (typ, performance mode)	0.007 °/s/ $\sqrt{\text{Hz}}$
Native ODR range	up to 6400 Hz

DepthAI-exposed runtime behaviour (this is what you actually get):

Parameter	Value
Stable accelerometer request rates	25 / 50 / 100 / 200 / 250 Hz
Stable gyroscope request rates	25 / 50 / 100 / 200 / 250 Hz
Practical maximum report rate	Requests above 400 Hz top out at ~250 Hz
Rate rounding	Requested rate rounds down to the next supported rate

Luxonis-characterized noise parameters (for filter tuning — units per Luxonis, treat as Allan-variance style parameters):

Axis	Accel noise density	Accel random walk	Accel bias stability	Gyro noise density	Gyro random walk	Gyro bias stability
X	0.05003	3.734e-5	0.06865	0.29334	0.28111	9.19266
Y	0.05473	5.082e-5	0.11015	0.28519	0.52435	9.68103
Z	0.06361	2.845e-5	0.04465	0.30118	0.25036	6.93407

Data received by host: `IMUData` messages containing one or more `IMUPacket` entries. Each packet exposes:

- `acceleroMeter` → `.x`, `.y`, `.z` as **float32, m/s²**, plus `.sequence` and `.timestamp`
- `gyroscope` → `.x`, `.y`, `.z` as **float32, rad/s**, plus `.sequence` and `.timestamp`

Report families (select at pipeline build time):

Suffix	Meaning
<code>_RAW</code>	Direct sensor output in the sensor-native frame
<code>_UNCALIBRATED</code>	Rotated into the Luxonis RDF frame using <code>imuExtrinsics</code> ; no IMU calibration applied
<code>_CALIBRATED</code>	Luxonis RDF frame and IMU calibration parameters applied

Batching is configurable via `setBatchReportThreshold(N)` and `setMaxBatchReports(M)` to trade host CPU load against latency.

Bandwidth @ 200 Hz, accel + gyro: ~6 floats + metadata ≈ 40 bytes × 200 ≈ **8 kB/s [EST]**

2.5 Output E — Neural Network Inference

Parameter	Value
Total VPU compute	4 TOPS
Neural inference allocation	1.4 TOPS
Model support	Any architecture, after conversion to the Myriad X blob format
Additional CV nodes	Warp/undistort, resize, crop (ImageManip), edge detection, feature tracking, custom CV
Object tracking	2D and 3D via ObjectTracker node

Data received by host: Structured metadata messages, not images —

- `ImgDetections` / `SpatialImgDetections`: list of detections, each with `label` (int), `confidence` (float 0-1), normalized bounding box (`xmin`, `ymin`, `xmax`, `ymax` as floats 0-1), and for spatial detections a `spatialCoordinates` field giving **X, Y, Z in millimetres** in the camera frame.
- `NNData`: raw output tensors for custom models.
- `Tracklets`: tracking ID, status (NEW/TRACKED/LOST/REMOVED), ROI, and 3D spatial coordinates.

This is the highest-value output for BillieBot — the device returns object identity *and* metric 3D position in one message, with no host-side depth lookup required.

3. Adafruit MLX90640 IR Thermal Camera Breakout — 55° (PID 4407)

Sensor: Melexis **MLX90640ESF-BAB-000-TU** (the **B** in the third option-code position denotes the 55° × 35° FOV lens) **Breakout physical:** 25.7 × 17.7 × 16.0 mm; STEMMA QT / Qwiic connectors plus breadboard-friendly headers **Package:** 4-lead TO-39 with integrated lens

3.1 Optical & Array

Parameter	Value
Array format	32 × 24 = 768 pixels (32 columns wide, 24 rows tall)
Field of view	55° (X / horizontal) × 35° (Y / vertical) , typical
Central pointing error from normal	±3° max
Instantaneous FOV per pixel (H)	~1.72° [CALC] (55° ÷ 32)
Instantaneous FOV per pixel (V)	~1.46° [CALC] (35° ÷ 24)
Ground sample at 1 m	~3.0 cm × 2.5 cm per pixel [CALC]
Ground sample at 3 m	~9.0 cm × 7.6 cm per pixel [CALC]
Defective pixels	Up to 4 permitted; flagged in on-chip EEPROM; replace by neighbour interpolation

BillieBot implication: At 3 m a standing adult (~50 cm wide) subtends only ~5-6 pixels horizontally. This sensor is a **presence/heat-signature detector**, not an imaging camera. Use it for human detection and hot-object avoidance, not for classification.

3.2 Measurement Performance

Parameter	Value
Target (object) temperature range	−40 °C to +300 °C
Ambient / operating temperature range	−40 °C to +85 °C
Stated accuracy (Adafruit, product level)	±2 °C in the 0–100 °C range
Frame accuracy (datasheet)	±1 °C (BAA reference case)
Non-uniformity, zone 1 (datasheet)	±0.5 °C (BAA reference case)
Total pixel absolute accuracy example (datasheet)	±1.5 °C for an 80 °C target in zone 1 (BAA)
NETD @ 1 Hz refresh, BAB variant	0.25 K average, 0.20 K minimum, $\sigma = 0.05$ K
NETD @ 1 Hz, BAA variant (for reference)	0.14 K average
ADC resolution	16 / 17 / 18 / 19-bit selectable; 18-bit default
Emissivity	User-supplied software parameter; not stored in the device

Accuracy caveats stated in the datasheet — all relevant to a mobile robot:

1. Accuracy specifications apply **only under settled isothermal conditions**.
2. Accuracy is valid **only when the target completely fills the pixel FOV**. A person at 3 m does not fill a pixel; expect the reported temperature to be a blend of the person and the background.
3. IR sensors are inherently susceptible to **thermal gradients**. Avoid mounting near motors, the VPU, or other heat sources.
4. Noise increases at lower target temperatures and decreases at higher ones.
5. Corner pixels are noisier than centre pixels due to lens optical performance.

3.3 Timing

Parameter	Value
Programmable refresh rate	0.5 / 1 / 2 / 4 / 8 / 16 / 32 / 64 Hz — default 2 Hz
Practical maximum (Adafruit)	~16 Hz (32 Hz is the theoretical limit but was not achievable in practice)
Subpage structure	The frame is split into 2 subpages . The configured refresh rate is the subpage rate.
Effective full-image rate	Half the configured refresh rate [CALC] — e.g. 16 Hz setting → 8 full thermal images/second
Reading pattern	Chess (default, factory-calibrated, recommended) or interleaved/TV mode
First valid data after power-on	40 ms + (1 / refresh rate) — e.g. 540 ms at the 2 Hz default
Thermal stabilization to rated accuracy	up to 4 minutes

Design note: The 4-minute thermal settling time means BillieBot's thermal channel should not be trusted for absolute temperature immediately after power-up. Either warm up before relying on absolute readings, or use relative/differential detection during the first several minutes.

3.4 Electrical

Parameter	Min	Typ	Max	Unit
Supply voltage (sensor, VDD)	3.0	3.3	3.6	V
Supply current (IDD)	14	18	25	mA
Power at 3.3 V, typ	—	~ 59.4 mW [CALC]	~82.5 mW [CALC]	—
Breakout board input voltage	3.0	—	5.0	V (on-board regulator + level shifting)
Breakout board total current	—	~20–25 mA [EST]	—	mA (sensor + regulator + pull-ups)
POR threshold (rising / falling)	2.2 / —	2.6 / 2.55	—	V

Melexis recommends holding VDD to **3.3 V ± 0.1 V** for best performance. Decoupling: 100 nF SMD plus 1 µF ceramic close to VDD/VSS, with short traces on **both** rails.

3.5 Digital Interface

Parameter	Value
Protocol	I²C , slave only
Default address	0x33 (programmable, up to 127 addresses; 0x00 must be avoided)
Clock frequency	400 kHz (standard/fast) up to 1 MHz (FM+ mode)
Logic levels	SDA/SCL are 5 V tolerant — can share a 5 V I ² C network directly
SDA sink current limit	10 mA (thermally limited; 20 mA is the I ² C spec ceiling)

3.6 Data Received from the Sensor

This is the most important section for BillieBot's software interface, because **the MLX90640 does not output temperature directly.**

What is physically on the wire (raw layer):

Item	Address	Format
IR pixel data	RAM 0x0400 - 0x06FF	768 × 16-bit signed (two's complement) ADC counts, proportional to received IR energy — not temperature
(Ta_Vbe)	RAM 0x0700	16-bit signed — ambient sensor
(CP) subpage 0	RAM 0x0708	16-bit signed — compensation pixel
(GAIN)	RAM 0x070A	16-bit signed
(Ta_PTAT)	RAM 0x0720	16-bit signed — ambient sensor
(CP) subpage 1	RAM 0x0728	16-bit signed
(VDD_pix)	RAM 0x072A	16-bit signed — supply sensor
Calibration constants	EEPROM 0x2400 - 0x273F	~832 words, read once after POR
Status register	0x8000	Bit 3 = "new data available in RAM" (host must clear it); bits 0-2 = last measured subpage
Control register 1	0x800D	Refresh rate, ADC resolution, reading pattern, subpage control

What the driver produces (application layer):

The host must (a) read the EEPROM calibration once after POR, (b) read both subpages, and (c) run the datasheet's compensation chain — supply voltage → ambient temperature → gain → offset/VDD/Ta compensation → emissivity → gradient → sensitivity normalization → **(To)**. The result is:

A 768-element array of IEEE-754 floating-point values in degrees Celsius, indexed row-major as a 32-wide × 24-tall image. Both the Adafruit CircuitPython and Arduino libraries return exactly this.

Pixel addressing: **(Pix(i, j))** where **(i)** = row 1-24 and **(j)** = column 1-32.

Bandwidth: 768 floats × 4 bytes × 8 full-frames/s ≈ **24.6 kB/s [CALC]** on the application side; ~13.3 kbit per subpage read on the I²C wire [CALC].

Extended-range detail: the compensation chain uses four object-temperature ranges with corner temperatures at -40 °C, 0 °C, CT3 (typ. 160 °C), and CT4 (typ. 320 °C), each with its own sensitivity slope **(KsTo)**. Standard driver libraries handle this transparently.

"Image mode" shortcut: If BillieBot only needs a relative thermal image (e.g. blob detection for a warm body) and not calibrated absolute temperature, the datasheet defines a reduced computation flow that skips the final $\overline{T_0}$ calculation — meaningfully lower host CPU cost.

4. Raspberry Pi Camera Module 3 NoIR — 75° (SC0873)

Sensor: Sony **IMX708**, back-illuminated stacked CMOS, Quad Bayer **Variant:** Standard lens (75° DFOV), **no IR cut filter**

4.1 Imaging

Parameter	Value
Resolution	11.9 MP
Pixel array	4608 × 2592
Pixel size	1.4 μm × 1.4 μm
Sensor diagonal	7.4 mm
Diagonal field of view	75°
Horizontal field of view	66°
Vertical field of view	41°
Focal length	4.74 mm
Focal ratio	f/1.8
Focus range	10 cm → ∞
Autofocus system	Phase Detection Autofocus (PDAF)
Shutter type	Rolling [EST] — not stated in the product brief; IMX708 is a rolling-shutter sensor
Infrared sensitive	Yes — no IR cut filter fitted
HDR mode	Yes, up to 3 MP output
Defect correction	Built-in 2D Dynamic Defect Pixel Correction (DPC)
Re-mosaic	QBC (Quad Bayer Coding) re-mosaic function

Common video modes (manufacturer-listed): 1080p50, 720p100, 480p120

BillieBot implication — NoIR colour behaviour: Without the IR cut filter, the Bayer colour filter array no longer isolates R/G/B cleanly; near-IR leaks into all three channels. Daylight images will appear reddish/washed out, and **colour-based classification will be unreliable**. Under IR illumination, treat this camera as effectively monochrome. Its value to BillieBot is night-vision and low-light operation, not colour fidelity.

4.2 Electrical & Environmental

Parameter	Value
Supply	3.3 V, sourced from the host's CSI connector
Current draw	~200-300 mA [EST] — Raspberry Pi does not publish this. Estimate derives from Raspberry Pi forum measurements of Camera Module 2/3 and the widely-cited 250 mA figure for Pi camera modules.
Power	~0.66 - 1.0 W [EST] (0.25 A × ~3.3 V, with margin)
Operating temperature	0 °C to 50 °C — the narrowest of any sensor in the suite; see §7.4
Dimensions	25 × 24 × 11.5 mm
Ribbon cable	200 mm, 15 × 1 mm FPC
Production lifetime	In production until at least January 2030

⚠ Verify current draw on the bench. This is the only significant power number in the suite that is not manufacturer-published. It matters because the CSI connector's 3.3 V rail is fed from the host SBC's regulator, which has a finite budget shared with other peripherals.

4.3 Digital Interface

Parameter	Value
Image data	MIPI CSI-2 serial output
Control	2-wire serial (I ² C), supports fast mode and fast-mode plus
Focus control	Separate 2-wire serial control of the focus mechanism
Software stack	<code>libcamera</code> / <code>picamera2</code> (fully supported, including autofocus). Not supported on Raspbian Buster or earlier.

4.4 Data Received from the Sensor

Layer	Format
Sensor output (on the CSI wire)	RAW10 — 10-bit Bayer, CSI-2 packed (<code>SRGBB10_CSI2P</code>), 5 bytes per 4 pixels
Post-ISP host formats	YUV420, NV12, RGB888, BGR888, XRGB8888 — selectable per-stream in <code>libcamera</code>
Application delivery	<code>picamera2</code> returns frames as NumPy arrays ; shape and dtype depend on the requested format (e.g. <code>(H, W, 3)</code> uint8 for RGB888)
Metadata per frame	Exposure time (μ s), analogue/digital gain, colour temperature, lens position (dioptries), focus FoM, sensor timestamp (ns)

Bandwidth:

- Full sensor read, 4608×2592 RAW10 ≈ 14.9 MB per frame [CALC]
- 1080p50 YUV420 = $1920 \times 1080 \times 1.5 \times 50 \approx \mathbf{155\ MB/s}$ [CALC] — host-side, post-ISP
- Common practice is the binned 2304×1296 mode for video, which halves the linear resolution and improves low-light SNR.

Autofocus data: `libcamera` exposes `AfState` (Idle / Scanning / Focused / Failed), `LensPosition` in reciprocal metres (dioptries), and `FocusFoM` (focus figure of merit). BillieBot can command `AfModeManual` with an explicit `LensPosition` to lock focus — recommended for a vibrating mobile platform, since continuous AF hunting wastes time and power.

5. Seed reSpeaker XVF3800 USB 4-Mic Array

Voice processor: XMOS **XVF3800** (xcore.ai XU316-1024-QF60B core), 7 × 7 mm 60-pin QFN **Audio codec:** Texas Instruments TLV320AIC3104 **Seed SKU:** 101991441 (variant without XIAO ESP32S3)

5.1 Acoustic Front End

Parameter	Value
Microphone count / type	4 × PDM MEMS, circular array
Pickup pattern	360°
Far-field pickup range	up to 5 m
AGC range	60 dB
Beamforming	Fast-tracking or fixed beamformer, multiple simultaneous beams
Acoustic Echo Cancellation	Full duplex AEC
Noise suppression	Dynamic / DNN-based echo and noise suppressor
De-reverberation	Yes
Voice Activity Detection	Yes
Direction of Arrival	Yes — see §5.4
Limiter	Yes
ASR output	Dedicated ASR beam with configurable fixed gain

5.2 Audio Data Format

Parameter	Value
Sample rate	16 kHz
Bit depth	32-bit (Seed firmware notes). Host stacks commonly negotiate <code>S32_LE</code> ; ALSA <code>plughw</code> will transparently convert to <code>S16_LE</code> if requested.
USB class	USB Audio Class 2.0 (high-speed USB 2.0 device)
Connector	USB Type-C (power + data)

Firmware-selectable channel maps:

Firmware	Channels	Channel assignment
<code>respeaker_xvf3800_usb_dfu_firmware_v2.0.x.bin</code>	2	Ch 0 = processed "Conference" audio; Ch 1 = processed "ASR" beam
<code>respeaker_xvf3800_usb_dfu_firmware_6chl_v2.0.x.bin</code>	6	Ch 0 = Conference; Ch 1 = ASR; Ch 2-5 = raw Mic 0-3

Recommendation for BillieBot: use the 6-channel firmware. It gives you the processed ASR beam for speech recognition *and* the four raw microphone streams, which you will need if you ever want to do your own beamforming, sound-source localization, or acoustic-event classification independent of the XMOS pipeline.

Bandwidth [CALC]:

- 2-channel: $16000 \times 4 \text{ bytes} \times 2 = \mathbf{128 \text{ kB/s}}$
- 6-channel: $16000 \times 4 \text{ bytes} \times 6 = \mathbf{384 \text{ kB/s}}$

Channel re-routing: `AUDIO_MGR_OP_L` / `AUDIO_MGR_OP_R` allow either output channel to be reassigned to a different internal source (e.g. amplified Mic 0, far-end reference).

5.3 Electrical

Parameter	Value
XVF3800 core (VDD) power, typical	400 mW in USB mode (345 mW in I ² S mode)
Core supply	0.9 V (VDD); 3.3 V (V_DDIOL/R/T); 1.8 V (VDD_IOB18); USB PHY 1.8 V + 3.3 V
Total board power	~1 - 2 W [EST] — Seeed does not publish a board-level figure. Estimate = 400 mW core + codec + regulators + PDM mics + USB PHY.
Total board current @ 5 V	~200 - 400 mA [EST]
Additional load: 12 × WS2812 RGB LEDs	up to ~0.7 A at full-white, full-brightness [EST] (12 × ~60 mA). At the moderate brightness used for DoA indication, budget ~60-150 mA [EST].
Additional load: JST speaker amplifier	Supports 5 W amplified speakers — if used, this dominates the board's power budget
Bus power	USB bus-powered; a USB 2.0 device may declare up to 500 mA

⚠ Power budgeting action: If BillieBot does not need the LED ring, **disable it** (`(led_effect 0)`, or drive GPO `(X0D33)` low to cut WS2812 power). This is the single largest discretionary load on this board. Similarly, hold GPO `(X0D31)` high to keep the speaker amplifier disabled if no speaker is fitted.

5.4 Control & Inference Data Interface

Control is available over **USB (EP0 vendor-specific class)** or **I²C**, via the `(xvf_host)` / `(xvf_host.exe)` utility or its underlying protocol. This is a **second, independent data path** from the audio stream.

Command	Returns	Format
<code>AEC_AZIMUTH_VALUES</code>	Direction of Arrival , 4 values	float32 radians (utility also prints degrees): [fixed beam 1, fixed beam 2, free-running beam, auto-selected beam]
<code>AEC_SPENERGY_VALUES</code>	Speech energy, 4 values	Integer energy per beam, same beam ordering
<code>GPI_READ_VALUES</code>	3 GPI pin states	3 × binary (e.g. <code>(1 0 0)</code>)
<code>GPO_READ_VALUES</code>	5 GPO pin states	5 × binary
<code>VERSION</code>	Firmware version	3 integers (e.g. <code>(2 0 2)</code>)

The auto-selected beam azimuth is the DoA value BillieBot should consume — it is the beam the pipeline chose for best audio, and it is what drives the LED ring indication.

GPIO map:

Pin	Dir	Function
<code>X1D09</code>	Input	Mute button status (high = released)
<code>X1D13</code> , <code>X1D34</code>	Input	Floating / available
<code>X0D30</code>	Output	Mute LED + microphone mute control (high = muted)
<code>X0D31</code>	Output	Amplifier enable (low = enabled)
<code>X0D33</code>	Output	WS2812 LED power control (high = on)
<code>X0D11</code> , <code>X0D39</code>	Output	Floating / available

Fixed-beam configuration (useful if BillieBot's microphone has a known preferred listening sector): `(AEC_FIXEDBEAMSAZIMUTH_VALUES <az1> <az2>)` (radians), `(AEC_FIXEDBEAMSELEVATION_VALUES <el1> <el2>)`, then `(AEC_FIXEDBEAMSONOFF 1)`.

Tuning parameters: `(AUDIO_MGR_REF_GAIN)`, `(AUDIO_MGR_MIC_GAIN)`, `(AUDIO_MGR_SYS_DELAY)`, `(PP_AGC_MAXGAIN)`, `(AEC_ASR_OUTGAIN)`.

5.5 Mechanical / Mounting Note

Seeed states the **microphone port (sound inlet) is on the back side of the board — the side with the printed Seeed Studio logo** — and must face the sound source for the acoustic

algorithms to work correctly. **This is a real integration constraint for BillieBot's enclosure design** and is easy to get backwards.

6. DFRobot Gravity: 10 DOF IMU AHRS — BNO055 + BMP280 (SEN0253)

Two independent sensors on one Gravity-I²C carrier board. They are addressed separately and are covered in §6.2 (BNO055) and §6.3 (BMP280).

6.1 Board-Level

Parameter	Value
Operating voltage	3.3 – 5 V DC
Operating current (DFRobot, board)	5 mA [CONFLICT]
BNO055 alone (Bosch, NDOF @ 100 Hz)	up to 12.3 mA [CONFLICT]
Interface	Gravity-I ² C (4-pin) plus a broken-out header
Operating temperature	–40 °C to +80 °C
Dimensions	32 × 27 mm
Board power @ 5 V	~25 mW (DFRobot figure) to ~62 mW [CALC] (Bosch figure)

[CONFLICT] — resolve on the bench. DFRobot's 5 mA board figure is **lower than Bosch's 12.3 mA maximum for the BNO055 alone**, before accounting for the BMP280, the level shifter, or pull-ups. The 5 mA figure is likely quoted for a low-power or non-fusion mode. **For BillieBot's power budget, use ~15–20 mA @ 5 V (~75–100 mW) as the conservative planning number [EST].** In absolute terms this is negligible next to the OAK-D Lite, so the discrepancy is low-risk — but do not use 5 mA in a battery-life calculation without verifying it.

Pinout:

#	Name	Function
1	VCC	Positive supply
2	GND	Ground
3	C	I ² C SCL
4	D	I ² C SDA
5	NBOOT	Boot mode select
6	RST	Reset
7	INT	Interrupt output
8	I2C_ADDR	BNO055 address select (0x28 / 0x29)
9	PS2	Protocol select 2
10	PS1	Protocol select 1
11	BL_IND	Bootstrap indication

Protocol selection (PS1 / PS2), default = 0 / 0:

PS1	PS2	Mode
0	0	Standard / Fast I²C (default)
0	1	HID over I ² C
1	0	UART
1	1	Reserved

I²C addresses: BNO055 = **0x28** (default; 0x29 selectable) · BMP280 = **0x76**

6.2 BNO055 — 9-Axis Absolute Orientation Sensor

Integrates a 14-bit triaxial accelerometer, a 16-bit triaxial gyroscope, a triaxial geomagnetic sensor, and a **32-bit Cortex-M0+ running Bosch sensor-fusion firmware**. The fusion is done on-chip; the host receives orientation directly.

6.2.1 Accelerometer

Parameter	Value
Ranges	$\pm 2 / \pm 4 / \pm 8 / \pm 16$ g (auto-set to ± 4 g in fusion modes)
Resolution	14-bit
Sensitivity	1 LSB/mg (all ranges)
Sensitivity tolerance	± 1 % typ, ± 4 % max @ 25 °C, ± 2 g
Zero-g offset	± 80 mg typ, ± 150 mg max over lifetime
Zero-g offset temp drift	± 1 mg/K typ, ± 3.5 mg/K max
Output noise density	150 $\mu\text{g}/\sqrt{\text{Hz}}$ typ , 190 $\mu\text{g}/\sqrt{\text{Hz}}$ max
Nonlinearity	0.5 % FS typ
Bandwidth options	7.81 / 15.63 / 31.25 / 62.5 (fusion default) / 125 / 250 / 500 / 1000 Hz
Cross-axis sensitivity	1 % typ, 2 % max
Package alignment error	0.5° typ, 2° max

6.2.2 Gyroscope

Parameter	Value
Ranges	$\pm 125 / \pm 250 / \pm 500 / \pm 1000 / \pm 2000$ °/s (auto-set to ± 2000 °/s in fusion)
Resolution	16-bit
Sensitivity	16.0 LSB/(°/s) , or 900 LSB/(rad/s)
Zero-rate offset	± 1 °/s typ, ± 3 °/s max
Zero-rate offset temp drift	± 0.015 °/s/K typ, ± 0.03 °/s/K max
Output noise	0.1 °/s rms typ (BW = 47 Hz), 0.3 °/s max $\rightarrow$ 0.014 °/s/ $\sqrt{\text{Hz}}$
Nonlinearity	± 0.05 % FS typ
Bandwidth options	12 / 23 / 32 (fusion default) / 47 / 64 / 116 / 230 / 523 Hz
Cross-axis sensitivity	± 1 % typ, ± 3 % max

6.2.3 Magnetometer

Parameter	Value
Range (X, Y)	$\pm 1300 \mu\text{T}$ typ ($\pm 1200 \mu\text{T}$ min)
Range (Z)	$\pm 2500 \mu\text{T}$ typ ($\pm 2000 \mu\text{T}$ min)
Device resolution	$0.3 \mu\text{T}$
Heading accuracy	$\pm 2.5^\circ$ typ, in a $30 \mu\text{T}$ horizontal geomagnetic field @ 25°C , fully calibrated with ideal tilt compensation
Zero-B offset (uncalibrated)	$\pm 40 \mu\text{T}$
Zero-B offset (after fusion-mode calibration)	$\pm 2 \mu\text{T}$
Zero-B offset temp drift	$\pm 0.23 \mu\text{T/K}$ typ, $\pm 0.37 \mu\text{T/K}$ max
Output noise (regular preset)	$0.6 \mu\text{T}$
Output noise (high-accuracy preset)	$0.3 \mu\text{T}$
Gain error after API compensation	$\pm 5\%$ typ, $\pm 8\%$ max
Resolution (x / y / z)	13 / 13 / 15 bits

6.2.4 Operating Modes & Output Rates

Mode	Accel	Mag	Gyro	Fusion Output	Fused Data Rate
ACCONLY / MAGONLY / GYROONLY	—	—	—	none	sensor-configured
ACCMAG / ACCGYRO / MAGGYRO / AMG	✓	✓	✓	none	sensor-configured
IMU	100 Hz	—	100 Hz	relative orientation	100 Hz
COMPASS	20 Hz	20 Hz	—	absolute	20 Hz
M4G	50 Hz	50 Hz	—	relative	50 Hz
NDOF_FMC_OFF	100 Hz	20 Hz	100 Hz	absolute	100 Hz
NDOF	100 Hz	20 Hz	100 Hz	absolute	100 Hz

- **NDOF** is the full 9-DOF absolute-orientation mode with Fast Magnetometer Calibration **on**. Highest accuracy and fastest magnetometer convergence; slightly higher current than **NDOF_FMC_OFF**.
- Data rate tolerance: $\pm 1\%$ using the internal oscillator (an external 32 kHz crystal improves this).

6.2.5 Electrical & Timing

Parameter	Value
VDD (sensors)	2.4 - 3.6 V
VDDIO (μ C and I/O)	1.7 - 3.6 V
Total supply current, normal mode, 9DOF @ 100 Hz	≤ 12.3 mA (VDD = 3 V, VDDIO = 2.5 V)
Total supply current, low-power mode	0.33 - 2.72 mA
Total supply current, suspend mode	≤ 0.04 mA
Operating temperature	-40 °C to +85 °C
Start-up time (off $\rightarrow$ CONFIG mode)	400 ms
POR time (reset $\rightarrow$ CONFIG mode)	650 ms
Mode switch, CONFIG $\rightarrow$ any operating mode	7 ms
Mode switch, any operating mode $\rightarrow$ CONFIG	19 ms
Mechanical shock survival	10,000 g (≤ 200 μ s) / 2,000 g (≤ 1 ms)

6.2.6 Data Received from the BNO055

All values are read from the register map as **signed integers**, and must be divided by the scaling factor below to obtain engineering units. Multi-byte reads (burst reads) are **register-shadowed**, guaranteeing LSB/MSB consistency — **single-byte reads are not, and can tear**. Always use burst reads.

Output	Registers	Size	Rate (NDOF)	Scaling	Units
Quaternion (w, x, y, z)	0x20 – 0x27	4 × int16	100 Hz	1 unit = 2¹⁴ = 16384 LSB	dimensionless
Euler angles (heading, roll, pitch)	0x1A – 0x1F	3 × int16	100 Hz	1° = 16 LSB or 1 rad = 900 LSB	° or rad
Accelerometer	0x08 – 0x0D	3 × int16	100 Hz	1 m/s² = 100 LSB , or 1 mg = 1 LSB	m/s ² or mg
Linear acceleration (gravity removed)	0x28 – 0x2D	3 × int16	100 Hz	1 m/s² = 100 LSB	m/s ² or mg
Gravity vector	0x2E – 0x33	3 × int16	100 Hz	1 m/s² = 100 LSB	m/s ² or mg
Gyroscope	0x14 – 0x19	3 × int16	100 Hz	1 °/s = 16 LSB , or 1 rad/s = 900 LSB	°/s or rad/s
Magnetometer	0x0E – 0x13	3 × int16	20 Hz	1 μT = 16 LSB	μT
Temperature	0x34	1 × int8	1 Hz	1 °C = 1 LSB (or 2 °F = 1 LSB)	°C or °F
Calibration status	0x35 (CALIB_STAT)	1 × uint8	on demand	2 bits each: system, gyro, accel, mag	0 (uncal) – 3 (fully cal)

Euler angle output ranges (Android format):

Angle	Range
Heading / Yaw	0° to 360° (clockwise increases)
Roll	–90° to +90°
Pitch	+180° to –180° (clockwise decreases)

Windows format is also selectable via `(UNIT_SEL)` bit 7, which changes the pitch sign convention. **Pick one and document it in BillieBot's ICD** — this is a classic source of sign-error bugs.

Unit selection register (`(UNIT_SEL)`, `0x3B`): acceleration m/s^2 vs mg · angular rate dps vs rps · Euler degrees vs radians · temperature $^{\circ}\text{C}$ vs $^{\circ}\text{F}$ · Windows vs Android orientation format.

Axis remapping: `(AXIS_MAP_CONFIG)` (`0x41`) and `(AXIS_MAP_SIGN)` (`0x42`) let the sensor's output frame be rotated in firmware to match BillieBot's body frame regardless of how the board is physically mounted. Eight standard placements (P0-P7) are tabulated in the datasheet. **Use this rather than rotating in host software** — it keeps the fusion output already in body frame.

Bandwidth [CALC]: reading quaternion + linear accel + gyro + calib status $\approx 21 \text{ bytes} \times 100 \text{ Hz} \approx 2.1 \text{ kB/s}$.

6.2.7 ⚠ Critical Application Warnings from the Bosch Datasheet

These are directly relevant to a wheeled/mobile robot and should drive BillieBot's sensor-fusion architecture:

1. **"The sensor fusion algorithm was primarily designed to track human motion. If the device is subjected to large accelerations for an extended period of time (e.g. in a vehicle cornering at high speed or braking over a long distance), the device may incorrectly interpret this large acceleration as the gravity vector."** → BillieBot's sustained accelerations during turns and braking may corrupt the roll/pitch estimate. **Test this specifically.**
2. **"The linear acceleration signal typically cannot be integrated to recover velocity, or double-integrated to recover position. The error typically becomes larger than the signal within less than 1 second"** unless other sensors compensate. → Do **not** dead-reckon from this IMU alone. Fuse with wheel odometry and/or the OAK-D Lite's visual data.
3. **Automatic background calibration cannot be disabled.** It runs continuously in all fusion modes.
4. Magnetometer distortion (from BillieBot's own motors, wiring, and battery currents) will cause the algorithm to ignore magnetometer data, at which point **heading drifts like IMU mode**. Mount the board as far from motors and high-current conductors as practical.
5. **Calibration procedure required after every power-on reset** unless a stored calibration profile is restored: accelerometer needs 6 stable orientations; gyroscope needs a few seconds stationary; magnetometer needs a figure-8 motion. Read `(CALIB_STAT)` to confirm. Calibration offsets/radii can be saved and re-written to skip this on subsequent boots.

6.3 BMP280 — Barometric Pressure & Temperature Sensor

Parameter	Value
Pressure range	300 - 1100 hPa
Relative accuracy	± 0.12 hPa ($\approx \pm 1$ m altitude)
Absolute accuracy	± 1 hPa ($\approx \pm 8.33$ m altitude)
Temperature range	0 °C to 65 °C
Temperature resolution	0.01 °C
I ² C address	0x76
Current draw	Not stated separately by DFRobot; negligible vs. the BNO055 [EST] (typically single-digit μ A at 1 Hz for this Bosch part)

Data received from the BMP280:

- **Raw layer:** 20-bit uncompensated pressure and 20-bit uncompensated temperature ADC values, plus a set of factory calibration coefficients in NVM. These raw values are **meaningless without compensation**.
- **Compensated layer (what the DFRobot library returns):**

Value	Units	Notes
Temperature	°C (float)	
Pressure	Pa (float)	Note: pascals , not hPa — the DFRobot example prints Pa
Altitude	m (float)	Calculated , not measured

BMP280 altitude caveat, stated explicitly by DFRobot: *"The altitude is calculated from the temperature and pressure values collected by the on-board sensor BMP280, not the actual measured value, and there is an error."* Altitude also requires a **sea-level pressure reference** (the DFRobot example hard-codes `SEA_LEVEL_PRESSURE 1015.0f`).

Absolute altitude is only as good as that reference. **Relative** altitude change (± 1 m) is far more trustworthy than absolute altitude (± 8.33 m).

BillieBot implication: For a ground robot, the useful signal here is **relative** — detecting that BillieBot has changed floors, gone up a ramp, or been picked up. Do not use

absolute barometric altitude for navigation. Also note the BMP280's **0–65 °C** temperature range is narrower than the BNO055's –40 to +85 °C.

7. System-Level Integration Notes

7.1 Consolidated Power Budget

Sensor	Rail	Typ. Current	Typ. Power	Peak Power	Provenance
OAK-D Lite (AF)	5 V USB	~0.8 – 1.0 A [CALC]	4 – 5 W	6 W	Stated
RPi Camera Module 3 NoIR	3.3 V CSI	~250 mA [EST]	~0.8 W [EST]	~1.0 W [EST]	Estimated
reSpeaker XVF3800 (LEDs off)	5 V USB	~200 – 400 mA [EST]	~1 – 2 W [EST]	~2 W [EST]	Core power stated; board power estimated
reSpeaker XVF3800 (LED ring on)	5 V USB	+ up to ~700 mA [EST]	+ up to ~3.5 W [EST]	—	Estimated
MLX90640 breakout	3.3 / 5 V	18 mA typ, 25 mA max	~60 mW [CALC]	~83 mW [CALC]	Stated (sensor)
SEN0253 (BNO055 + BMP280)	3.3 / 5 V	~15 mA [EST]	~75 mW [EST]	~100 mW [EST]	See §6.1 conflict
TOTAL (LEDs off)	—	—	≈ 6 – 8 W [CALC]	≈ 9 – 10 W [CALC]	—
TOTAL (LEDs on)	—	—	≈ 9 – 11 W [CALC]	≈ 13 W [CALC]	—

Recommendation: Size BillieBot's sensor power rail for $\geq 15\text{ W}$ to preserve margin. The OAK-D Lite dominates — it is ~60–70 % of the sensor suite's draw on its own.

7.2 ⚠ USB Current Constraint — OAK-D Lite

Luxonis states directly: *"Consumption of OAK-D Lite can be higher and will fall out of the USB2 maximum specified range of 900 mA."* On hosts with limited per-port current (Raspberry Pi 4 and similar), Luxonis supplies a **Y-adapter** that brings in a separate power feed alongside the host data connection.

Action for BillieBot: Either (a) power the OAK-D Lite through a Y-adaptor from BillieBot's main rail, or (b) use a powered USB hub, or (c) confirm the host SBC can source ≥ 1.2 A on that single port. **Do not assume the SBC's USB port can carry it.** Undervoltage on this device manifests as intermittent enumeration failures and pipeline crashes, which are painful to diagnose.

7.3 ⚠️ I²C Bus Speed Conflict — MLX90640 vs. BNO055

All three I²C devices have distinct addresses (0x33, 0x28, 0x76) and *can* electrically share one bus. **But they should not.**

Device	Max I ² C clock
MLX90640	1 MHz (FM+) — and it <i>needs</i> the speed
BNO055	Standard / Fast I ² C (400 kHz)
BMP280	400 kHz typical

A shared bus must run at the slowest device's rate: **400 kHz**.

Bandwidth analysis at 400 kHz [CALC]:

- One MLX90640 subpage read $\approx 768 \times 2$ bytes + auxiliary registers ≈ 1664 bytes ≈ 13.3 kbit
- At 400 kHz with I²C overhead: $\approx$ **40–45 ms per subpage**
- Two subpages per full thermal image: $\approx$ **80–90 ms** $\rightarrow$ **~ 11 full images/second maximum**, *with the bus doing nothing else*
- Add the BNO055 at 100 Hz (~ 2.1 kB/s) and the bus contention gets tight.

Recommendation: Put the **MLX90640 on its own I²C bus at 1 MHz**, and the BNO055 + BMP280 together on a second bus at 400 kHz. Most SBCs expose multiple I²C controllers. If only one bus is available, cap the MLX90640 at **8 Hz refresh (4 full images/second)** to leave headroom. Also note the MLX90640 datasheet's warning that **capacitive loading degrades I²C** — keep the bus short and consider stronger pull-ups.

7.4 Environmental Envelope — Limiting Component

Sensor	Operating Temperature Range
MLX90640	−40 °C to +85 °C
BNO055	−40 °C to +85 °C
SEN0253 board	−40 °C to +80 °C
OAK-D Lite	−20 °C to +50 °C
RPi Camera Module 3	0 °C to +50 °C ← limiting component
BMP280 (temperature measurement range)	0 °C to +65 °C

BillieBot's usable ambient envelope is 0 °C to +50 °C, set by the Camera Module 3. Raspberry Pi additionally warns the camera should be operated in a **well-ventilated environment** and that a case should not be covered — relevant if the camera is enclosed in BillieBot's shell. The OAK-D Lite's −20/+50 °C figure applies at *full VPU utilization*; running lighter pipelines raises the tolerable ambient.

7.5 Aggregate Data Bandwidth [CALC]

Stream	Rate
OAK-D Lite RGB, 1080p30 uncompressed NV12	~93 MB/s
OAK-D Lite RGB, 1080p30 H.265	~1-2 MB/s [EST]
OAK-D Lite mono ×2, 640×480 @ 30 fps	~18.4 MB/s
OAK-D Lite depth, 640×480 uint16 @ 30 fps	~18.4 MB/s
OAK-D Lite IMU @ 200 Hz	~8 kB/s
OAK-D Lite NN detections	< 10 kB/s [EST]
RPi Camera 3, 1080p50 YUV420	~155 MB/s
reSpeaker, 6-channel	384 kB/s
MLX90640 @ 8 full images/s	~24.6 kB/s
SEN0253 @ 100 Hz	~2.1 kB/s

Observation: The two cameras dominate by three orders of magnitude. **Use on-device H.265/MJPEG encoding on the OAK-D Lite and keep the Pi camera in a binned mode** unless full-resolution uncompressed frames are genuinely required. All the low-rate sensors combined (IMU, thermal, audio, AHRS) total well under 0.5 MB/s and are effectively free.

7.6 Complementary Coverage — Field of View Comparison

Sensor	HFOV	VFOV	DFOV
OAK-D Lite RGB	69°	54°	81°
OAK-D Lite mono / depth	73°	58°	86°
RPi Camera 3 NoIR	66°	41°	75°
MLX90640 (55°)	55°	35°	~65° [CALC]
reSpeaker XVF3800	360°	—	—

The four optical sensors have broadly comparable and overlapping fields of view, with the **MLX90640 the narrowest** — if the thermal channel is to be co-registered with the RGB or depth image, the thermal FOV defines the usable common region. The microphone array is the only sensor with full 360° awareness, making its DoA output BillieBot's primary cue for **directing attention outside the optical field of view**.

8. Open Items / Bench Verification Required

#	Item	Section	Why it matters
1	Confirm the OAK-D Lite unit has an IMU (Kickstarter units do not)	§2.4	An entire data stream may not exist
2	Characterize actual usable stereo depth range	§2.3	Official sources conflict (80 cm–12 m vs 40 cm–8 m)
3	Measure RPi Camera Module 3 current draw	§4.2	Only unpublished significant power number
4	Measure reSpeaker XVF3800 board current, LEDs on and off	§5.3	Board-level power not published
5	Measure SEN0253 current in NDOF mode	§6.1	DFRobot (5 mA) vs Bosch (12.3 mA) conflict
6	Test BNO055 fusion under sustained cornering/braking	§6.2.7	Datasheet warns of gravity-vector misinterpretation
7	Characterize magnetometer distortion from BillieBot's motors	§6.2.7	Determines whether absolute heading is usable at all
8	Decide I ² C bus topology (split vs shared) and verify MLX90640 throughput	§7.3	Sets the achievable thermal frame rate
9	Verify USB power delivery path for the OAK-D Lite	§7.2	Undervoltage causes hard-to-diagnose failures
10	Confirm reSpeaker microphone-inlet orientation in the enclosure	§5.5	Acoustic algorithms depend on it

9. Sources

Luxonis OAK-D Lite

- Luxonis hardware documentation — OAK-D Lite: <https://docs.luxonis.com/hardware/products/OAK-D%20Lite>
- Luxonis OAK-D Lite datasheet (Dec 2021), electrical characteristics and camera sensor characteristics
- Luxonis DepthAI IMU node documentation: <https://docs.luxonis.com/software-v3/depthai/depthai-components/nodes/imu.md>

- Luxonis BMI270 hardware reference:
<https://docs.luxonis.com/hardware/platform/sensors/imu/bmi270.md>
- Luxonis store page — OAK-D Lite: <https://shop.luxonis.com/products/oak-d-lite-1>

Adafruit MLX90640 (PID 4407)

- Adafruit product page: <https://www.adafruit.com/product/4407>
- Melexis MLX90640 datasheet, Revision 11, 4 May 2018:
<https://www.melexis.com/-/media/files/documents/datasheets/mlx90640-datasheet-melexis.pdf>

Raspberry Pi Camera Module 3 NoIR (SC0873)

- Raspberry Pi Camera Module 3 product brief, June 2024:
<https://datasheets.raspberrypi.com/camera/camera-module-3-product-brief.pdf>
- Raspberry Pi product page: <https://www.raspberrypi.com/products/camera-module-3/>
- Raspberry Pi camera documentation:
<https://www.raspberrypi.com/documentation/accessories/camera.html>
- Current-draw estimate: Raspberry Pi community forum measurements (see §4.2)

Seeed reSpeaker XVF3800

- Seeed Studio Wiki — Getting Started with reSpeaker XVF3800:
https://wiki.seeedstudio.com/respeaker_xvf3800_introduction/
- XMOS XVF3800 datasheet v3.2.1, Key Features (power consumption):
https://www.xmos.com/documentation/XM-014888-PC/html/modules/fw_k_xvf/doc/datasheet/01_features.html
- XMOS XVF3800 datasheet v3.2.1, Device Operation (power supplies):
https://www.xmos.com/documentation/XM-014888-PC/html/modules/fw_k_xvf/doc/datasheet/06_device_operation.html
- reSpeaker XVF3800 firmware and host control repository:
https://github.com/respeaker/respeaker_XVF3800_USB_4MIC_ARRAY

DFRobot SEN0253

- DFRobot Wiki — Gravity: 10 DOF IMU AHRS BNO055 + BMP280:
<https://wiki.dfrobot.com/sen0253/>
- Bosch Sensortec BNO055 datasheet, BST-BNO055-DS000-18, Revision 1.8, October 2021: <https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bno055-ds000.pdf>
- DFRobot BMP280 example code and library documentation:
<https://wiki.dfrobot.com/sen0253/docs/23846>

